Shihan Sharar

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Education

University of Waterloo — *Waterloo, ON*
Bachelor of Computer Science — *Sept 2023 - Apr 2028 (Expected)*

- **Dean's List** (all terms); CGPA: **94%**, Faculty GPA: **96%** (with advanced courses)
- **Clubs:** Measuring rover inclination angle for UW Robotics Club, member of Waterloo Quant and UW Putnam Club
- **Skills**: C++, C, C#, Python, Java, Go, Ruby, Rust, React, TypeScript/JavaScript, Node.js

Experience

Persona — *San Francisco, California*
Software Engineer Intern - Platform Experience — *June 2026 - Present*

- Re-architected dashboard event retrieval in **Ruby on Rails** and **TypeScript** behind a unified server-side API, implementing cursor pagination over a hybrid **MongoDB**/**Snowflake** query layer with permission-aware serialization and batched actor/resource hydration.
- Built runtime infrastructure for Layout Editor computed properties, executing **JSON-Logic** and sandboxed **QuickJS** custom code in CSV export pipelines to enforce dynamic visibility rules outside the browser.
- Developing a **Cloudflare** AI UI-authoring **eval** harness with parallel **Cursor Agent SDK** runs across prompt, model, and skill configs, artifact capture, LLM-as-judge scoring, persisted runs for skill comparisons, and deterministic **TSX AST** scorers for design-system usage and code-style violations.

University of Waterloo — *Waterloo, ON*
Compiler Research Assistant - Prof. Y. Zhang — *Dec 2025 - Present*

- Built a **Python** compiler pass for a probabilistic programming language (**PPL**) that lowers continuous distributions into fixed-point Boolean arithmetic for **JAX/XLA** inference, generating incremental-carry addition, bitwise comparisons, and per-bit conditioning.
- Implemented bit-blasting for **exponential**, **Laplace**, and **gamma** distributions, and extended the compiler to arbitrary densities using **CDF**-fitted exponential segments with interleaved variable elimination to bound tensor size.

BitGo — *Palo Alto, California*
Software Engineer Intern - Prime Trade — *Sep 2025 - Dec 2025*

- Designed and engineered an **LP-settlement** tracking system in **Go**, introducing new data models in **PostgreSQL**, **GraphQL** schemas/resolvers, and APIs to track settlement lifecycle state and reconcile trades.
- Developed an ingestion framework for **Talos** OTC fills, producing normalized data for the settlement system.
- Built **React** dashboards with filtering and sorting over ingestion and settlement data, cutting operational settlement workflows by ~**20 minutes** per case; validated with **Vitest**.
- Implemented a notification pipeline, defining **gRPC** schemas, producers that write to an **outbox**, a publisher that streams events to **Kafka**, and consumers that trigger **Novu** workflows for over **10,000** notifications daily.
- Eliminated duplicate notifications and hardened delivery across failures using deterministic **idempotency** keys, persisted delivery state, and exponential-backoff retries with a DLQ.

University of Waterloo — *Waterloo, ON*
System Security Research Assistant - Prof. N. Asokan — *May 2025 - Aug 2025*

- Engineered a **C++** command-line interface using the **llama.cpp** API, adding an executor framework that maps model outputs to deterministic system actions with full control-flow tracing.
- Built a secure-world **OP-TEE** monitor in **C** (emulated ARM TrustZone) to validate all LLM interactions via signing/hashing, preventing indirect prompt-injection and enforcing control-flow integrity (**CFI**).

Projects

Velocore: Multithreaded C++ Trading Engine

- Built a multithreaded trading engine using **Crow**, **Boost.Asio**, & **Boost.Beast** for async I/O, & websocket communication; parsing **1,000+** orders in under **30ms** with low-latency trade execution and health-check endpoints.
- Integrated **Alpaca's** market-data feed and paper-trading **REST API** into a custom broker interface for real-time quote ingestion, order management, and portfolio tracking.